AUDIT ANALYTICS FINDINGS REPORT

Amani Microfinance Ltd

Reporting Period: Financial Year 2025
Prepared by: Audit Analytics System
Report Generated: 30 June 2026

This report presents the results of an automated audit analytics review covering transaction-level anomaly detection and loan portfolio reconciliation procedures. Findings herein are intended to support, not replace, professional auditor judgment and further investigation.

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Executive Summary

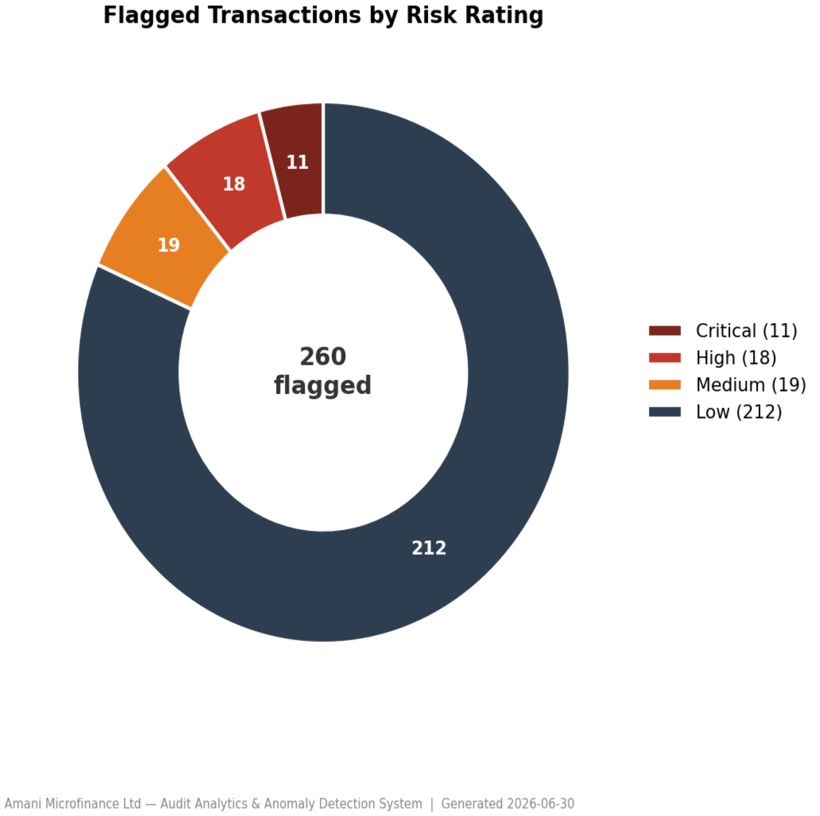
This engagement applied automated audit analytics procedures to 496 general ledger transactions and 37 active loan accounts for Amani Microfinance Ltd. Procedures performed included Benford's Law digit-frequency analysis, duplicate payment detection, round-number transaction screening, robust statistical outlier detection (modified z-score and IQR methods), and reconciliation of general ledger loan balances against the loan portfolio management system's supporting schedule. Individual findings were combined into a single weighted risk score per transaction, reflecting both the strength of corroborating evidence and the financial materiality of each item.

Key Findings at a Glance

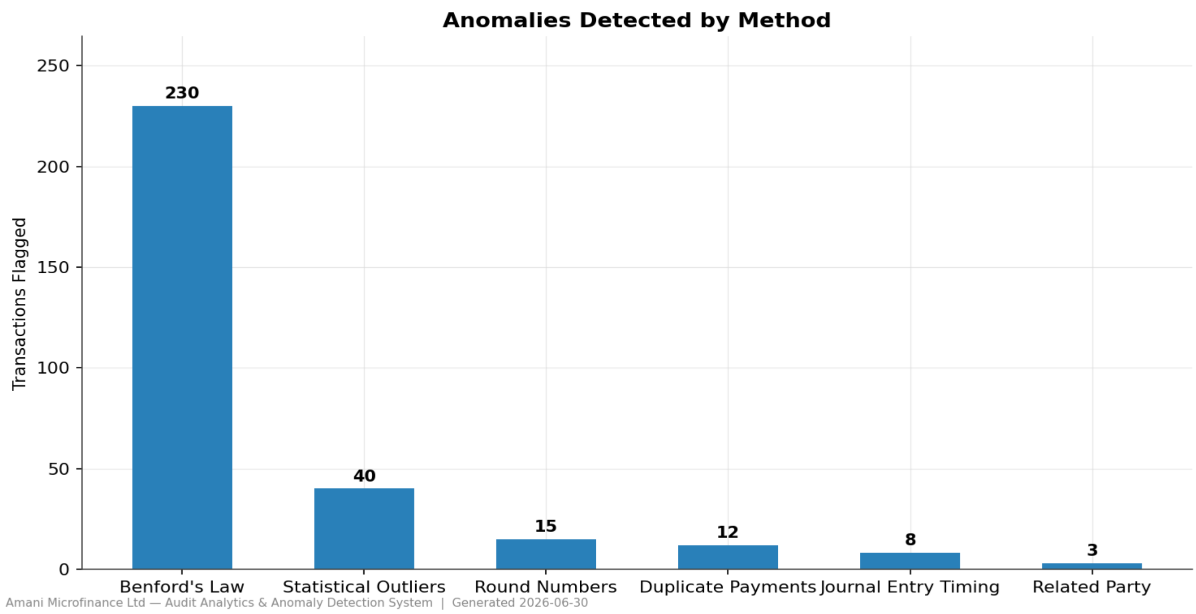
Metric	Value
Total transactions reviewed	496
Transactions flagged by at least one procedure	260
Critical-risk transactions	11
High-risk transactions	18
Total exposure — Critical & High risk transactions (TZS)	44,112,400
Loan accounts reviewed for reconciliation	37
Reconciliation exceptions requiring follow-up	10
Possible ghost loans identified (no GL support)	2

Priority attention required: 11 transaction(s) reached the highest (Critical) risk rating, and 2 loan account(s) on the active portfolio schedule have no corresponding general ledger activity — a pattern consistent with fictitious or unrecorded lending. Both categories warrant immediate management follow-up ahead of detailed substantive testing.

Risk Profile of Flagged Transactions

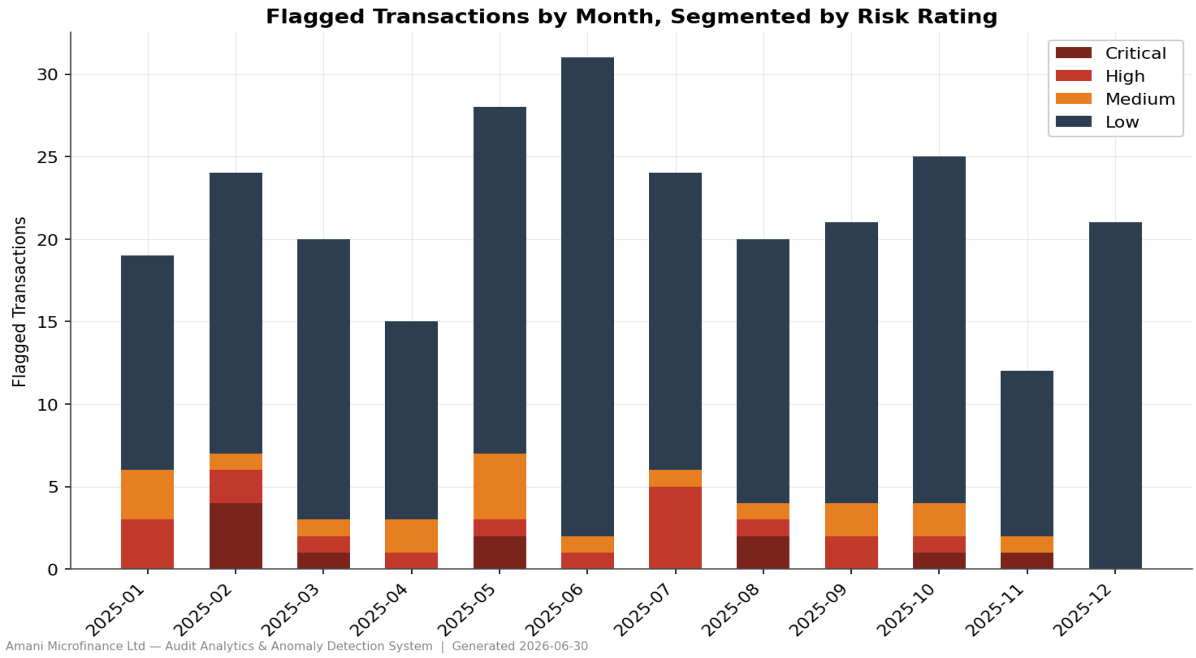


Anomalies Detected by Method



Reading this chart: Benford's Law operates as a population-level statistical test, flagging any transaction whose leading digit falls within an over-represented bucket. This makes it intentionally broad — a high count from this method alone reflects statistical breadth, not confirmed wrongdoing. Duplicate payment detection, by contrast, uses strict matching logic and produces far fewer but substantially higher-confidence results. The weighted risk score (see following pages) accounts for this difference in evidentiary strength; this chart should not be read as a ranking of method importance.

Flagged Transactions Over Time



Top 10 Highest-Risk Transactions

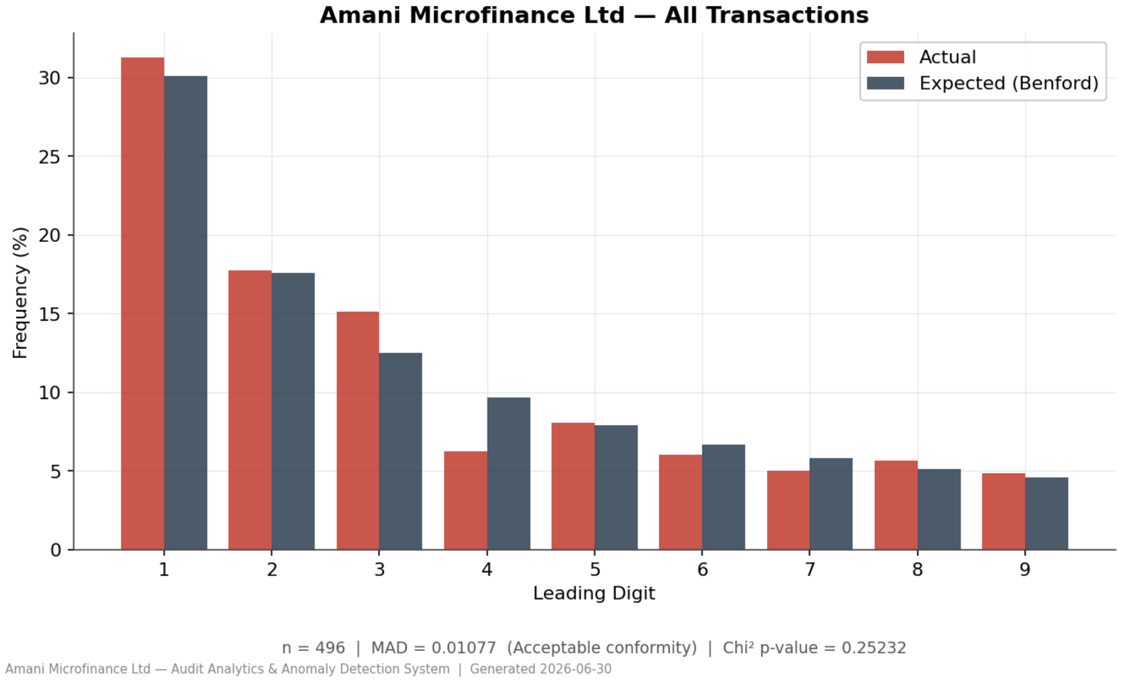
Transaction ID	Date	Amount (TZS)	Account	Risk Score	Rating
TXN10006	2025-02-10	390,000	Savings Deposit	100.0	Critical
TXN-DUP4	2025-02-10	390,000	Savings Deposit	100.0	Critical
TXN-DUP2	2025-05-21	373,900	Savings Deposit	93.8	Critical
TXN10071	2025-05-21	373,900	Savings Deposit	93.8	Critical
TXN-RPT2	2025-02-19	2,929,200	Loan Disbursement	87.5	Critical
TXN-DUP5	2025-08-05	444,000	Loan Loss Provision	81.2	Critical
TXN10031	2025-08-05	444,000	Loan Loss Provision	81.2	Critical
TXN-RND6	2025-03-12	3,000,000	Loan Disbursement	68.8	Critical
TXN-RND7	2025-10-06	3,000,000	Loan Disbursement	68.8	Critical
TXN-RND0	2025-11-19	10,000,000	Loan Disbursement	68.8	Critical

Detailed Findings

1. Benford's Law Digit-Frequency Analysis

Benford's Law predicts the expected frequency of leading digits in naturally occurring financial data, following a logarithmic rather than uniform distribution. Deviations can indicate estimated, manually adjusted, or fabricated entries. Conformity was assessed using both a chi-square goodness-of-fit test and Nigrini's Mean Absolute Deviation (MAD), the standard forensic-accounting metric for this test.

Sample size (transactions tested)	496
Chi-square statistic (p-value)	10.184 (0.25232)
Mean Absolute Deviation (MAD)	0.01077
Conformity rating	Acceptable conformity



2. Duplicate Payment Detection

Transactions sharing the same counterparty, account, and amount within a 3-day window were flagged as potential duplicate payments — a pattern consistent with accidental double-disbursement or processing error, and representing direct financial exposure rather than a purely statistical indicator.

Transaction ID	Date	Amount (TZS)	Account	Counterparty	Group
TXN10411	2025-03-26	24,500	Savings Deposit	Grace Mboya	DUP-GRP-1
TXN-DUP0	2025-03-26	24,500	Savings Deposit	Grace Mboya	DUP-GRP-1
TXN10062	2025-10-08	10,700	Interest Income	Grace Shayo	DUP-GRP-2
TXN-DUP3	2025-10-08	10,700	Interest Income	Grace Shayo	DUP-GRP-2
TXN10031	2025-08-05	444,000	Loan Loss Provision	Hamisi Kileo	DUP-GRP-3

Transaction ID	Date	Amount (TZS)	Account	Counterparty	Group
TXN-DUP5	2025-08-05	444,000	Loan Loss Provision	Hamisi Kileo	DUP-GRP-3
TXN10406	2025-07-22	26,000	Savings Deposit	Hamisi Shayo	DUP-GRP-4
TXN-DUP1	2025-07-22	26,000	Savings Deposit	Hamisi Shayo	DUP-GRP-4
TXN10071	2025-05-21	373,900	Savings Deposit	Joseph Lyimo	DUP-GRP-5
TXN-DUP2	2025-05-21	373,900	Savings Deposit	Joseph Lyimo	DUP-GRP-5
TXN10006	2025-02-10	390,000	Savings Deposit	Lucy Ngonyani	DUP-GRP-6
TXN-DUP4	2025-02-10	390,000	Savings Deposit	Lucy Ngonyani	DUP-GRP-6

3. Round-Number Transaction Screening

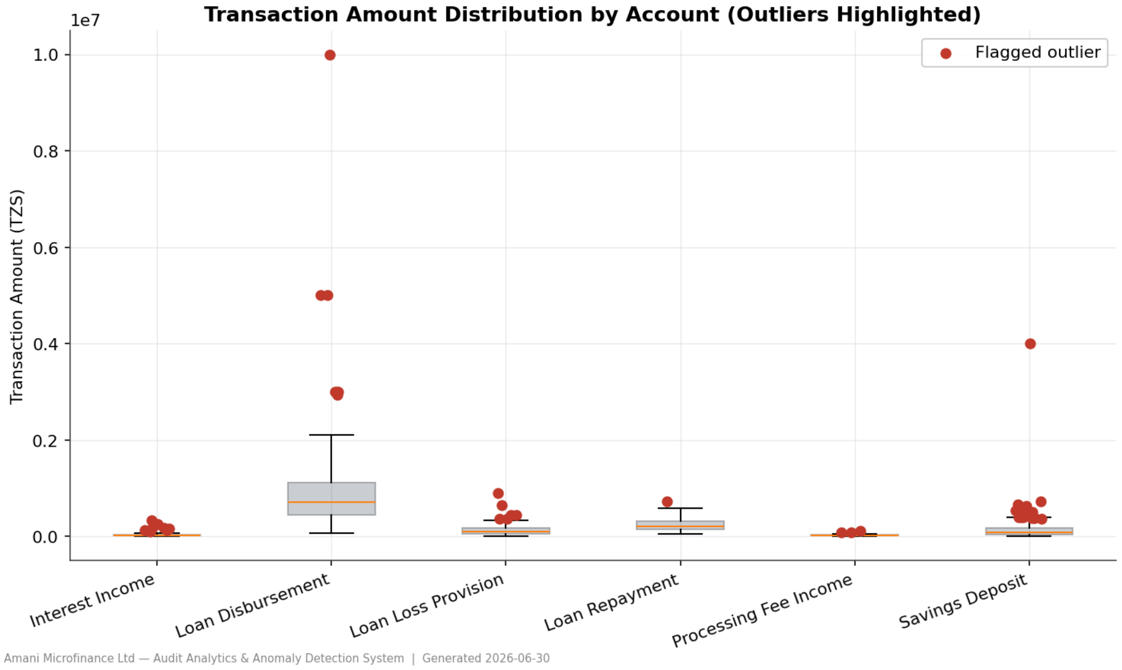
Transactions with amounts that are exact multiples of round figures (TZS 10,000 and above) were flagged on a tiered basis — the rounder the amount, the higher the indicator weight. Genuine transaction amounts rarely land on perfectly round figures by chance; round amounts are more often associated with estimates, manual journal entries, or fabricated values, though legitimate standardized loan products can also produce round disbursements.

Transaction ID	Date	Amount (TZS)	Account	Roundness Tier
TXN-RND0	2025-11-19	10,000,000	Loan Disbursement	Extremely round (multiple of 1,000,000)
TXN-RND2	2025-06-19	5,000,000	Loan Disbursement	Extremely round (multiple of 1,000,000)
TXN-RND3	2025-07-22	5,000,000	Loan Disbursement	Extremely round (multiple of 1,000,000)
TXN-RND7	2025-10-06	3,000,000	Loan Disbursement	Extremely round (multiple of 1,000,000)
TXN-RND1	2025-02-03	3,000,000	Loan Disbursement	Extremely round (multiple of 1,000,000)
TXN-RND6	2025-03-12	3,000,000	Loan Disbursement	Extremely round (multiple of 1,000,000)
TXN-RND8	2025-06-26	2,000,000	Loan Disbursement	Extremely round (multiple of 1,000,000)
TXN-RND9	2025-01-14	1,000,000	Loan Disbursement	Extremely round (multiple of 1,000,000)
TXN-RND4	2025-05-21	1,000,000	Loan Disbursement	Extremely round (multiple of 1,000,000)
TXN-RND5	2025-07-30	500,000	Loan Disbursement	Very round (multiple of 500,000)
TXN10006	2025-02-10	390,000	Savings Deposit	Moderately round (multiple of 10,000)
TXN-DUP4	2025-02-10	390,000	Savings Deposit	Moderately round (multiple of 10,000)

Transaction ID	Date	Amount (TZS)	Account	Roundness Tier
TXN10137	2025-10-17	160,000	Loan Repayment	Moderately round (multiple of 10,000)
TXN10003	2025-04-23	100,000	Interest Income	Highly round (multiple of 100,000)
TXN10069	2025-08-15	40,000	Savings Deposit	Moderately round (multiple of 10,000)

4. Statistical Outlier Detection

Transaction amounts were tested for statistical outliers WITHIN each account type using the modified z-score method (median and median absolute deviation), which is robust to the skewed distributions typical of financial data and is not self-distorted by the outliers it detects — unlike the classic mean/standard-deviation z-score, which this engagement also computed for comparison but did not rely upon as the primary method.

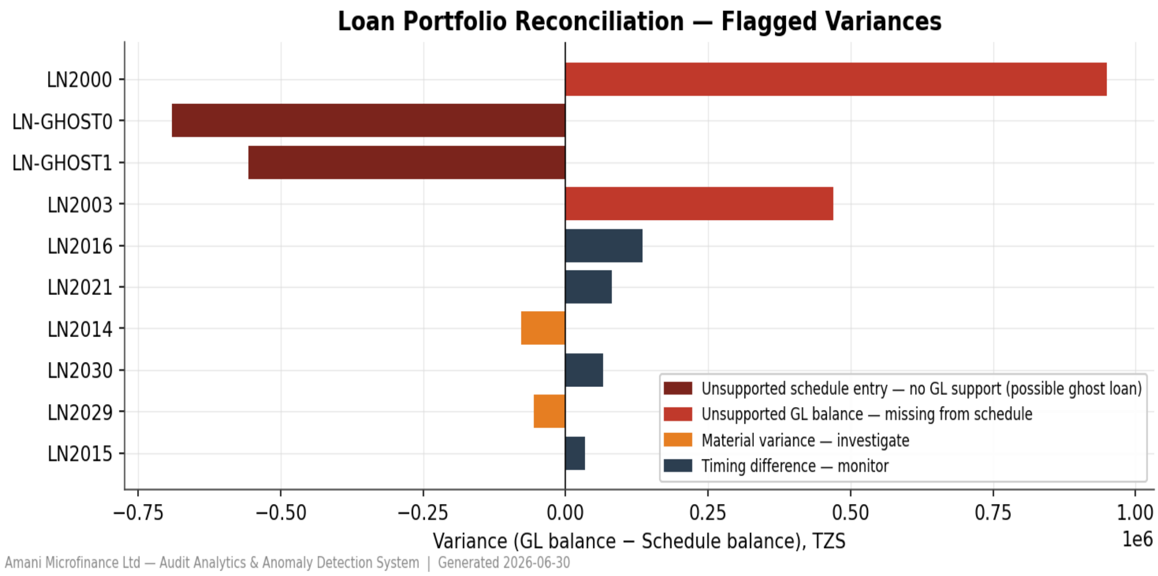


Flagged Outliers by Account

Account	Outliers Flagged
Savings Deposit	14
Interest Income	9
Loan Disbursement	7
Loan Loss Provision	6
Processing Fee Income	3
Loan Repayment	1

5. Loan Portfolio Reconciliation

General ledger loan balances (disbursements less repayments, by loan account) were reconciled against the loan portfolio management system's supporting schedule. Each loan was classified as a clean tie-out, a timing difference (consistent with one unposted installment), a material variance, an unsupported GL balance (present in the GL but missing from the schedule), or an unsupported schedule entry (present on the schedule with no corresponding GL activity — a pattern consistent with a fictitious or unrecorded loan). Materiality thresholds were applied on a percentage basis given the wide range of loan sizes in the portfolio.



All balances in Tanzanian Shillings (TZS).

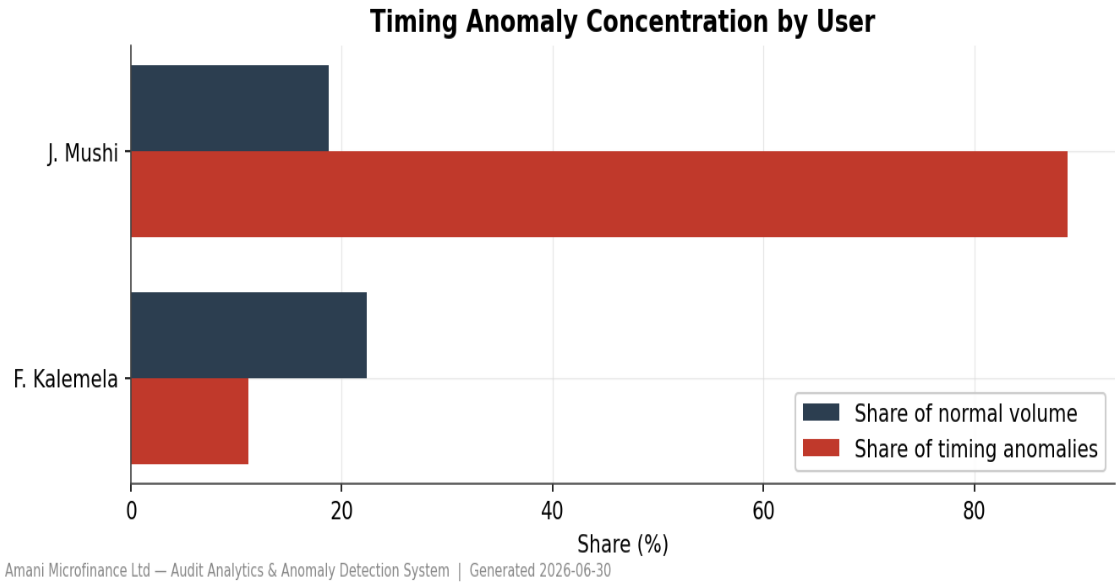
Loan ID	Borrower	GL Balance	Schedule Balance	Variance	Category
LN-GHOST0	Catherine Shayo	—	691,000	—	**Unsupported schedule entry**
LN-GHOST1	Elisha Mwakatobe	—	556,000	—	**Unsupported schedule entry**
LN2000	Baraka Kileo	950,300	—	—	**Unsupported GL balance**
LN2003	Pendo Shayo	469,400	—	—	**Unsupported GL balance**
LN2014	Pendo Ngonyani	131,500	209,357	-77,857	**Material variance**
LN2029	Catherine Mwakatobe	119,500	175,958	-56,458	**Material variance**
LN2016	Salome Lyimo	949,600	813,900	135,700	**Timing difference**
LN2021	Grace Massawe	653,200	571,500	81,700	**Timing difference**
LN2030	Joseph Kway	398,800	332,400	66,400	**Timing difference**
LN2015	Catherine Mrema	269,800	236,000	33,800	**Timing difference**

6. Journal Entry Testing (Timing & User Concentration)

Entries posted on weekends or outside standard business hours (08:00-18:00) were identified, then cross-referenced against each user's overall posting volume to compute a concentration ratio — the user's share of timing-anomalous entries divided by their share of normal activity. A ratio near 1.0 indicates proportionate, unremarkable activity; a ratio well above 1.0 indicates a user disproportionately associated with off-hours postings, which can be consistent with bypassing normal segregation-of-duties oversight. Only users with both an elevated ratio and a minimum number of flagged entries are escalated, to avoid flagging isolated, innocent after-hours activity.

Total timing-anomalous entries	**9**
Weekend postings	4
After-hours postings	5

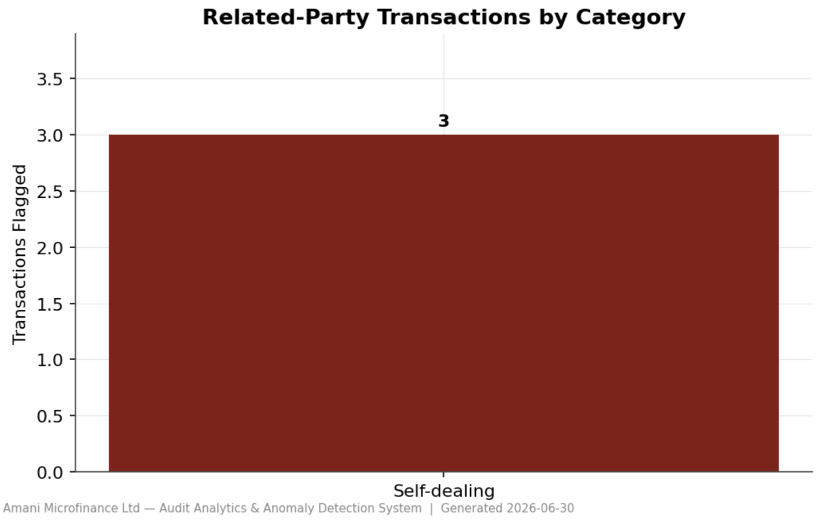
Disproportionate concentration identified: J. Mushi — see chart and table below for the specific concentration ratios driving this finding.



User	Total Txns	% of Volume	Flagged Txns	% of Anomalies	Concentration
J. Mushi	93	18.8%	8	88.9%	4.74x
F. Kalemela	111	22.4%	1	11.1%	0.5x

7. Related-Party Transaction Screening

Counterparty (borrower) names were compared against the roster of staff members appearing as transaction posters. A match does not by itself indicate impropriety — many institutions legitimately extend staff loan benefit schemes — but it creates a disclosure and scrutiny obligation under IAS 24 (Related Party Disclosures) and ISA 550 (Auditing Related Party Relationships and Transactions). The highest-risk pattern is specifically **self-dealing**: a staff member appearing as the counterparty on a transaction THEY THEMSELVES posted, which bypasses the basic control that origination and approval be performed by different people.



Transaction ID	Date	Amount (TZS)	Posted By	Counterparty	Category
TXN-RPT0	2025-01-02	532,500	A. Mwakasege	A. Mwakasege	**Self-dealing**
TXN-RPT2	2025-02-19	2,929,200	F. Kalemela	F. Kalemela	**Self-dealing**
TXN-RPT1	2025-12-05	344,200	A. Mwakasege	A. Mwakasege	**Self-dealing**

Appendix: Methodology & Limitations

Scope of Procedures

This report was generated using an automated audit analytics toolkit applying seven procedures: Benford's Law digit-frequency analysis, duplicate payment detection, round-number transaction screening, robust statistical outlier detection, journal entry testing (timing and user concentration analysis), related-party transaction screening, and general ledger-to-schedule reconciliation of the loan portfolio. Individual findings were combined into a weighted risk score reflecting both the strength of corroborating evidence across methods and the financial materiality of each transaction.

Known Limitations

- Benford's Law is a population-level statistical test, not a transaction-level proof of wrongdoing. Reliability also depends on sample size — segments below approximately 300 observations should be treated as indicative only, per Nigrini's published guidance.
- Duplicate payment detection depends on the accuracy and consistency of the counterparty field in the source data. Inconsistent name recording could cause genuine duplicates to be missed (a false negative risk).
- Round-number screening is a risk indicator, not proof of fabrication. Legitimate standardized loan products can produce coincidentally round disbursement amounts.
- Statistical outlier detection compares transactions within their own account type to avoid cross-contamination between accounts of different typical scale. Results depend on each account having a reasonably sized and representative population of normal transactions.
- Journal entry testing assumes a standard 08:00-18:00 business-hours window and a Monday-Friday working week; institutions with shift-based operations, multiple time zones, or extended service hours would require these parameters to be recalibrated. A user's concentration ratio reflects correlation with off-hours activity, not direct proof of wrongdoing — legitimate explanations (different working patterns, remote disbursement duties) should be considered before escalating.
- Related-party screening relies on the staff roster derived from the 'posted by' field and exact (or closely similar) name matching against the counterparty field. It cannot detect relationships not evidenced by name similarity — for example, a relative or associate of staff with a different surname — and a name match does not by itself establish impropriety; legitimate staff loan benefit schemes are common at financial institutions.
- The weighted risk-scoring weights applied in this engagement (related-party self-dealing, duplicate payment, journal entry timing concentration, statistical outlier, round-number tier, and Benford indicators) are calibrated illustratively for this engagement and should be reviewed and adjusted by the engagement team based on entity-specific risk factors before relying on them for a real client.
- Loan portfolio reconciliation procedures were applied to Loan Disbursement and Loan Repayment general ledger activity only. Other account types (Savings Deposits, Interest Income, Processing Fee Income, Loan Loss Provision) were not subject to reconciliation procedures in this engagement and would require separate substantive testing.

- All findings in this report are generated by automated procedures and are intended to direct and prioritize professional audit attention — they do not constitute audit conclusions and do not replace the exercise of professional judgment, corroborating inquiry, or further substantive testing.